

Refine judgement, checking, and mathematical communication

Solve each problem carefully. Check your work, decide if your answer makes sense, and explain your thinking clearly.

Step 12

Upper Elementary

★ Today I am learning to refine judgement, checking, and mathematical communication.

1. Check for Reasonableness

Solve each problem. Then check your answer. Decide if it is reasonable.

- a) A store sold 248 water bottles in the morning and 326 in the afternoon. How many in all?



My answer: _____

Check (show your work):

Is my answer reasonable?

☐ Yes ☐ No Why? _____

- b) A school has 1,245 students. If 678 are girls, how many are boys?



My answer: _____

Check (show your work):

Is my answer reasonable?

☐ Yes ☐ No Why? _____

- c) A baker used 3.6 kg of flour in the morning and 2.85 kg in the afternoon. How many kilograms in all?



My answer: _____

Check (show your work):

Is my answer reasonable?

☐ Yes ☐ No Why? _____

- d) A book costs \$14.75. A notebook costs \$3.80. If I buy 2 books and 3 notebooks, how much will I pay?



My answer: \$ _____

Check (show your work):

Is my answer reasonable?

☐ Yes ☐ No Why? _____

2. Explain Your Thinking

Solve the problem. Then explain how you solved it.

- a) $732 + 298 =$ _____

Explain your thinking.



I used this strategy: _____
I decided it was the best because: _____

- b) $1,452 - 678 =$ _____

Explain your thinking.



I used this strategy: _____
I decided it was the best because: _____

3. Choose and Justify

For each problem, choose the best strategy. Then justify your choice.

Problem	Strategy Choices (choose one)	My Choice	Why is this the best strategy?
a) $598 + 276$	☐ Mental math ☐ Standard algorithm ☐ Compensation	☐	_____
b) $3.75 + 4.8 + 2.25$	☐ Grouping ☐ Estimation ☐ Standard algorithm	☐	_____
c) 856×25	☐ Break apart ☐ Use partial products ☐ Repeated addition	☐	_____
d) $\$85.60 - \37.45	☐ Estimate first ☐ Standard algorithm ☐ Round and adjust	☐	_____

4. Communicate Clearly

Write a clear solution to each problem. Show your work and use words.

- a) A theatre sold 356 tickets on Friday, 489 on Saturday, and 278 on Sunday. How many tickets did it sell in all over the weekend?

Solution:

Answer: _____

- b) A recipe needs 1.25 cups of sugar. Julia makes 3 batches of the recipe. How many cups of sugar does she need in total?

Solution:

Answer: _____



- c) A bike is \$246.50. A helmet costs \$38.75. If the store is having a 15% discount on the total, how much will I pay?

Solution:



Answer: \$ _____

5. Error Analysis

Each student solved the problem below. Find and explain the mistake. Then solve it correctly.

Problem: $724 + 389 = ?$

Sam's Work

$$\begin{array}{r} 724 \\ + 389 \\ \hline 1013 \end{array}$$

What is the mistake?

Correct solution:

Maya's Work

$$\begin{array}{r} 1 1 \\ 724 \\ + 389 \\ \hline 1113 \end{array}$$

What is the mistake?

Correct solution:

6. Reflect and Improve

Answer the questions below.

- What steps do you take to check your answers? _____
- How do you decide if an answer is reasonable? _____
- Which strategy helps you most often? Why? _____
- What can you do when you get stuck on a problem? _____
- What is one thing you will improve next time? _____



My Learning Check

Circle how confident you feel.



I can do it!



I need more practice.



I need help.

I used...



checking



explaining



choosing strategies



clear communication



judgement

Today I learned...

1 thing I did well: _____

1 thing I will improve: _____